MATH 3350-014 Exam II-B 3 April 2012

Make-Up

Name ________________________ Score ____________________

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** Attach this sheet to the front of your answers.

1. (8 pts) Find the general solution of the differential equation $2y'' - 6y' - 5y = 0$

2. (8 pts) Find the general solution of the differential equation $4y'' - 6y' + 5y = 0$

3. (14 pts) Solve the initial-value problem $y'' - 6y' + 25y = 0$, $y(0) = -5$, $y'(0) = -2$

4. (14 pts) Find the general solution of the differential equation $y'' - 2y' - 15y = 4 - 75x^2$

5. (14 pts) Find the general solution of the differential equation $y'' - 9y = 6e^{-3x} + 7 - 4x$

6. (8 pts) Find the general solution of the differential equation $2x^2y'' - 11xy' + 11y = 0$

7. (8 pts) Find the general solution of the differential equation $2x^2y'' - 6xy' + 12y = 0$

8. (8 pts) Solve the initial-value problem $x^2y'' + 6xy' - 24y = 0$, $y(1) = -7$, $y'(1) = 12$

9. (14 pts) Use the method of *Variation of Parameters* to find a particular solution of $y'' - 6y' + 9y = e^{3x} \ln x$

10. (10 pts) A mass weighing 9 lbs is attached to a 12 ft spring and stretches the spring so that its new length is 15 ft. This mass is replaced by another mass which weighs 24 lbs. The medium through which the spring-mass system moves creates a damping force equal to one and a half the instantaneous velocity. The mass is released at a point ½ ft below the equilibrium position with an initial upward velocity of 1 ft/s. Find the equation of motion.